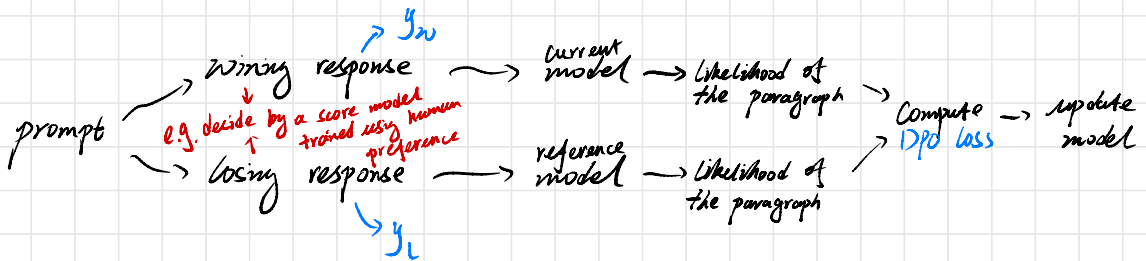


Direct Preference Optimization (DPO) is a method for aligning LLM with human preference, such as being helpful, positive, harmless etc.

In RLHF, we build a separate Value model to address the credit assignment problem. In DPO, we frame the alignment problem as a simple Classification problem.

The idea of DPO is quite simple



DPO loss:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\log \left(\beta \log \frac{\pi_{\theta}(y_w)}{\text{ref}(y_w)} - \beta \log \frac{\pi_{\theta}(y_l)}{\text{ref}(y_l)} \right)$$

Optimize θ to minimize $\mathcal{L}$

Current model

reference model:
frozen during DPO

$$\log(\pi_{\theta}(y_w)) = \sum_i \log(\pi_{\theta}(y_{w, \text{token } i} | x, y_{w, \text{token } 1:i-1}))$$

$\log \frac{\pi_{\theta}(y_w)}{\text{ref}(y_w)}$: measures how much more likely the new model to generate the winning response than losing response

$\log \frac{\log(y_i)}{\log(y_i)}$: measures how much more likely the new model to generate the losing response than winning response

$\log \frac{\log(y_w)}{\log(y_w)} - \log \frac{\log(y_i)}{\log(y_i)}$: measures the gap of preference for y_w and y_i

β : Controls how much we want to update the new model compared with the reference model

if β is large, then $\sigma(\beta(\cdot)) \rightarrow 1$, the

loss $\rightarrow 0$, so the loss will be small.

Comparison with RLHF:

	RLHF (PPO)	DPO
number of models involved	4 → score model → value model ↓ → reference model new model	2 → reference model → new model
training method	RL	Supervised learning
gradient taken	token level	Sequence level
Learning style	Online (needs to generate)	Offline
Cost	high	low
Performance	better (if train stably...)	lower (but training is easier)